```
import seaborn as sns
# Load Titanic dataset
df = sns.load_dataset("titanic")
df.head()


# sns.get_dataset_names()
```

Show hidden output

```
import seaborn as sns
# Load Titanic dataset
df = sns.load_dataset("titanic")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
import seaborn as sns
# Load Titanic dataseti
df = sns.load_dataset("titanic")
df.describe()
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
import seaborn as sns
# Load Titanic dataseti
df = sns.load_dataset("titanic")
df.columns
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

```
df. isnull()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
887	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
888	False	False	False	True	False	False	False	False	False	False	False	True	False	False	False
889	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False

891 rows × 15 columns

```
import pandas as pd

data = {
    "name": ["Alice", "Bob", "Charlie"],
    "age": [25, 30, 35],
    "city": ["NYC", "LA", "Chicago"]
}

df10 = pd.DataFrame(data)

print(df)
```

```
     survived  pclass     sex   age  sibsp  parch     fare embarked   class  \
0           0       3    male  22.0      1      0   7.2500        S   Third
1           1       1  female  38.0      1      0  71.2833        C   First
2           1       3  female  26.0      0      0   7.9250        S   Third
3           1       1  female  35.0      1      0  53.1000        S   First
4           0       3    male  35.0      0      0   8.0500        S   Third
..        ...     ...     ...   ...    ...    ...      ...      ...     ...
886         0       2    male  27.0      0      0  13.0000        S  Second
887         1       1  female  19.0      0      0  30.0000        S   First
888         0       3  female   NaN      1      2  23.4500        S   Third
889         1       1    male  26.0      0      0  30.0000        C   First
890         0       3    male  32.0      0      0   7.7500        Q   Third

       who  adult_male deck  embark_town alive  alone
0      man        True  NaN  Southampton    no  False
1    woman       False    C    Cherbourg   yes  False
2    woman       False  NaN  Southampton   yes   True
3    woman       False    C  Southampton   yes  False
4      man        True  NaN  Southampton    no   True
..     ...         ...  ...          ...   ...    ...
886    man        True  NaN  Southampton    no   True
887  woman       False    B  Southampton   yes   True
888  woman       False  NaN  Southampton    no  False
889    man        True    C    Cherbourg   yes   True
890    man        True  NaN   Queenstown    no   True

[891 rows x 15 columns]
```

```
df.isnull()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False
887	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
888	False	False	False	True	False	False	False	False	False	False	False	True	False	False	False
889	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False	False	False	True	False	False	False

891 rows × 15 columns

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
df2=df.dropna()
df2
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alon
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fals
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fals
6	0	1	male	54.0	0	0	51.8625	S	First	man	True	E	Southampton	no	Tru
10	1	3	female	4.0	1	1	16.7000	S	Third	child	False	G	Southampton	yes	Fals
11	1	1	female	58.0	0	0	26.5500	S	First	woman	False	C	Southampton	yes	Tru
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	.
871	1	1	female	47.0	1	1	52.5542	S	First	woman	False	D	Southampton	yes	Fals
872	0	1	male	33.0	0	0	5.0000	S	First	man	True	B	Southampton	no	Tru
879	1	1	female	56.0	0	1	83.1583	C	First	woman	False	C	Cherbourg	yes	Fals
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	Tru
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	Tru

182 rows × 15 columns

```
df.dropna(subset=["age"])
```

```
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	T
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	T
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	T
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	T
888	0	3	female	NaN	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	T
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	T

891 rows × 15 columns

```
df["age"]=df["age"].fillna(0)
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	Tı
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	Tı
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	Tı
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	Tı
888	0	3	female	0.0	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	Tı
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	Tı

891 rows × 15 columns

```
df["fare"] =df["fare"].fillna(df["fare"].mean())
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	Tı
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	Tı
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	Tı
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	Tı
888	0	3	female	0.0	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	Tı
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	Tı

891 rows × 15 columns

```
df.isnull().sum()/len(df)*100
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	Tı
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	Tı
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	Tı
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	Tı
888	0	3	female	0.0	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	Tı
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	Tı

891 rows × 15 columns

```
df['age'].mean()
```

```
np.float64(23.79929292929293)
```

```
df["age"].mean()
```

```
np.float64(23.79929292929293)
```

```
df["age"]=df["age"].fillna(df["age"].mean())
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	Tı
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	Tı
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	Tı
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	Tı
888	0	3	female	0.0	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	Tı
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	Tı

891 rows × 15 columns

```
df['fare'].mean()
```

```
np.float64(32.204207968574636)
```

```
df["age"]=df["age"].fillna(df["fare"].mean())
df
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alo
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fa
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	Tı
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	Tı
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
886	0	2	male	27.0	0	0	13.0000	S	Second	man	True	NaN	Southampton	no	Tı
887	1	1	female	19.0	0	0	30.0000	S	First	woman	False	B	Southampton	yes	Tı
888	0	3	female	0.0	1	2	23.4500	S	Third	woman	False	NaN	Southampton	no	Fa
889	1	1	male	26.0	0	0	30.0000	C	First	man	True	C	Cherbourg	yes	Tı
890	0	3	male	32.0	0	0	7.7500	Q	Third	man	True	NaN	Queenstown	no	Tı

891 rows × 15 columns

```
df.drop(["alive", "who", "embarked","deck"], axis=1, inplace=True)
df
```

	survived	pclass	sex	age	sibsp	parch	fare	class	adult_male	embark_town	alone
0	0	3	male	22.0	1	0	7.2500	Third	True	Southampton	False
1	1	1	female	38.0	1	0	71.2833	First	False	Cherbourg	False
2	1	3	female	26.0	0	0	7.9250	Third	False	Southampton	True
3	1	1	female	35.0	1	0	53.1000	First	False	Southampton	False
4	0	3	male	35.0	0	0	8.0500	Third	True	Southampton	True
...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	Second	True	Southampton	True
887	1	1	female	19.0	0	0	30.0000	First	False	Southampton	True
888	0	3	female	0.0	1	2	23.4500	Third	False	Southampton	False
889	1	1	male	26.0	0	0	30.0000	First	True	Cherbourg	True
890	0	3	male	32.0	0	0	7.7500	Third	True	Queenstown	True

891 rows × 11 columns

```
# df=df[["alone","age","survived","fare","class"]]
# df
```

```
df["sex_code"] = df["sex"].map({"male":10 ,"female": 9})
df
```

	survived	pclass	sex	age	sibsp	parch	fare	class	adult_male	embark_town	alone	sex_code
0	0	3	male	22.0	1	0	7.2500	Third	True	Southampton	False	10
1	1	1	female	38.0	1	0	71.2833	First	False	Cherbourg	False	9
2	1	3	female	26.0	0	0	7.9250	Third	False	Southampton	True	9
3	1	1	female	35.0	1	0	53.1000	First	False	Southampton	False	9
4	0	3	male	35.0	0	0	8.0500	Third	True	Southampton	True	10
...	...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	Second	True	Southampton	True	10
887	1	1	female	19.0	0	0	30.0000	First	False	Southampton	True	9
888	0	3	female	0.0	1	2	23.4500	Third	False	Southampton	False	9
889	1	1	male	26.0	0	0	30.0000	First	True	Cherbourg	True	10
890	0	3	male	32.0	0	0	7.7500	Third	True	Queenstown	True	10

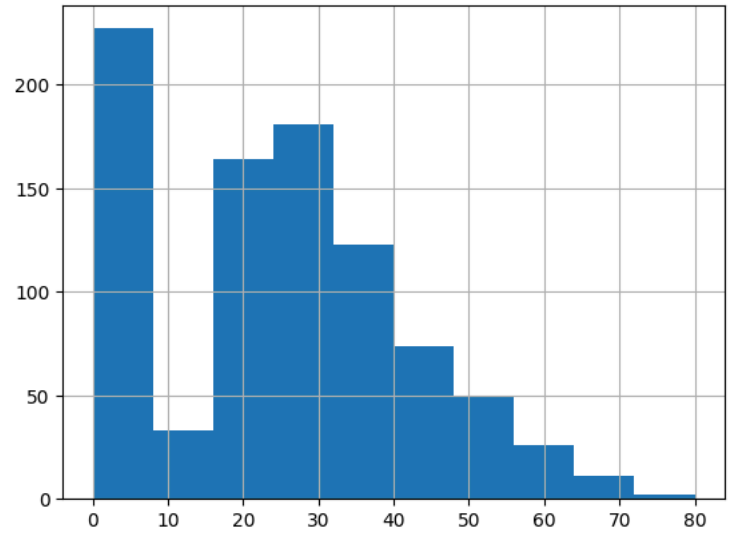
891 rows × 12 columns

```
df[["pclass", "survived"]].corr()
```

	pclass	survived
pclass	1.000000	-0.338481
survived	-0.338481	1.000000

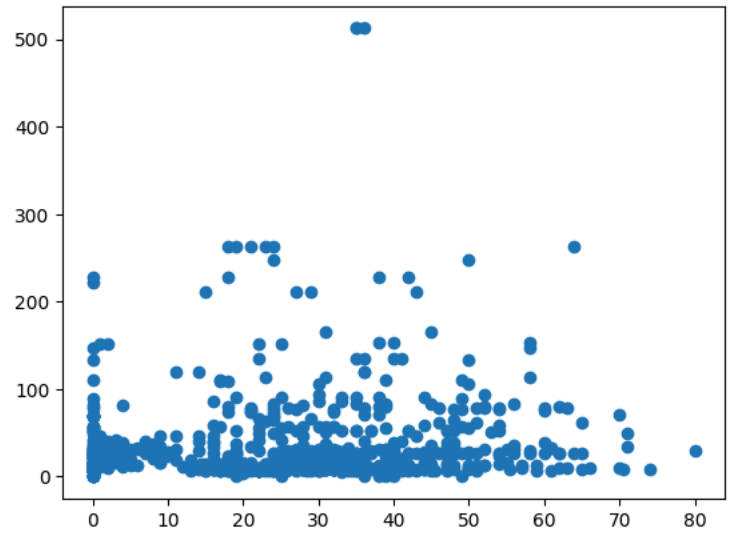
```
df["age"].hist()
```

```
<Axes: >
```




```
import matplotlib.pyplot as plt
plt.scatter(df["age"], df["fare"])
```

```
<matplotlib.collections.PathCollection at 0x7b61e04b0cb0>
```




```
df["fare"].min()
```

```
0.0
```

```
df["fare"].max()
```

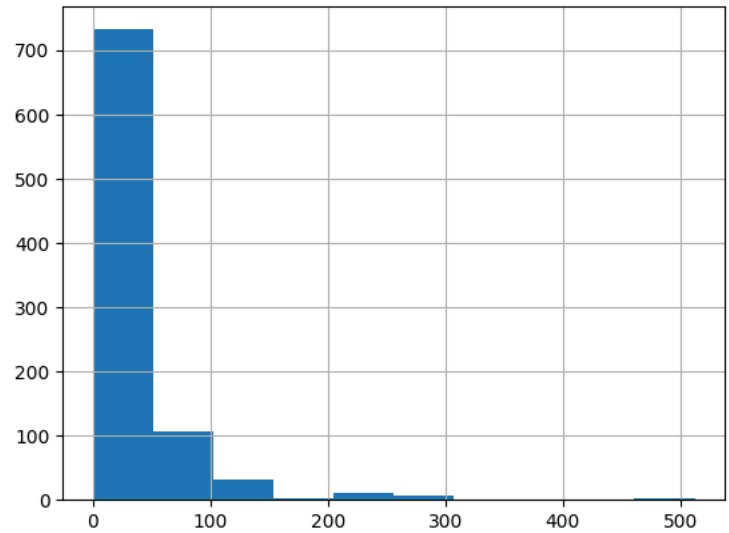
```
512.3292
```

```
df["fare"].mean()
```

```
np.float64(32.204207968574636)
```

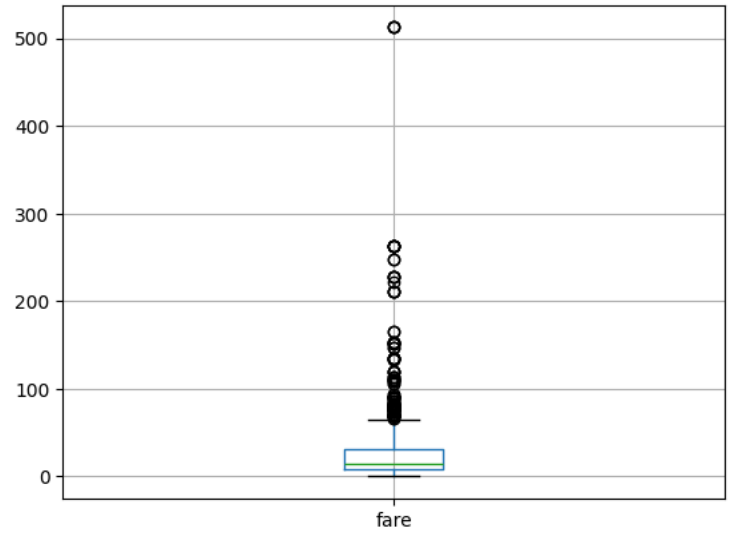
```
df["fare"].hist()
```

```
<Axes: >
```




```
import matplotlib.pyplot as pltplt
df.boxplot(column="fare")
```

```
<Axes: >
```




```
q1=df["fare"].quantile(0.25)
q1
```

```
np.float64(7.9104)
```

```
q3=df["fare"].quantile(0.75)
q3
```

```
np.float64(31.0)
```

```
q2=df['fare'].quantile(0.50)
q2
```

```
np.float64(14.4542)
```

```
iqr=q3-q1
iqr
```

```
np.float64(23.0896)
```

```
lower=q1-1.5*iqr
lower
```

```
np.float64(-26.724)
```

```
upper=q3+1.5*iqr
upper
```

```
np.float64(65.6344)
```

```
outlier=df[df["fare"]<lower]
outlier
```

survived	pclass	sex	age	sibsp	parch	fare	class	adult_male	embark_town	alone	sex_code

```
outlier=df[df["fare"]>upper]
outlier
```

	survived	pclass	sex	age	sibsp	parch	fare	class	adult_male	embark_town	alone	sex_code
1	1	1	female	38.0	1	0	71.2833	First	False	Cherbourg	False	9
27	0	1	male	19.0	3	2	263.0000	First	True	Southampton	False	10
31	1	1	female	0.0	1	0	146.5208	First	False	Cherbourg	False	9
34	0	1	male	28.0	1	0	82.1708	First	True	Cherbourg	False	10
52	1	1	female	49.0	1	0	76.7292	First	False	Cherbourg	False	9
...	...	...	...	...	...	...	...	...	...	...	...	...
846	0	3	male	0.0	8	2	69.5500	Third	True	Southampton	False	10
849	1	1	female	0.0	1	0	89.1042	First	False	Cherbourg	False	9
856	1	1	female	45.0	1	1	164.8667	First	False	Southampton	False	9
863	0	3	female	0.0	8	2	69.5500	Third	False	Southampton	False	9
879	1	1	female	56.0	0	1	83.1583	First	False	Cherbourg	False	9

116 rows × 12 columns

```
outlier.shape
```

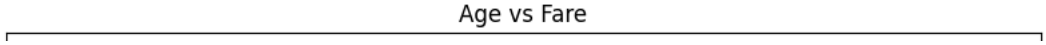
```
(116, 12)
```

```
dfnooutlier =df[(df["fare"] >= lower)&(df["fare"] <=upper)]
dfnooutlier
```

	survived	pclass	sex	age	sibsp	parch	fare	class	adult_male	embark_town	alone	sex_code
0	0	3	male	22.0	1	0	7.2500	Third	True	Southampton	False	10
2	1	3	female	26.0	0	0	7.9250	Third	False	Southampton	True	9
3	1	1	female	35.0	1	0	53.1000	First	False	Southampton	False	9
4	0	3	male	35.0	0	0	8.0500	Third	True	Southampton	True	10
5	0	3	male	0.0	0	0	8.4583	Third	True	Queenstown	True	10
...	...	...	...	...	...	...	...	...	...	...	...	...
886	0	2	male	27.0	0	0	13.0000	Second	True	Southampton	True	10
887	1	1	female	19.0	0	0	30.0000	First	False	Southampton	True	9
888	0	3	female	0.0	1	2	23.4500	Third	False	Southampton	False	9
889	1	1	male	26.0	0	0	30.0000	First	True	Cherbourg	True	10
890	0	3	male	32.0	0	0	7.7500	Third	True	Queenstown	True	10

775 rows × 12 columns

```
import matplotlib.pyplot as plt

df = sns.load_dataset("titanic")
```

```
plt.figure(figsize=(10, 6))
plt.plot(df["age"], df["fare"])

plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```